# ONUR AYTEKIN

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## EDUCATION

**Air Force Institute of Technology (AFIT), Wright-Patterson AFB, OH** Expected March 2027
***M.S., Astronautical Engineering — GPA 3.98/4.0***

- **Thesis:** Reinforcement learning for Space Situational Awareness (SSA) sensor-network tasking, incorporating maneuvering spacecraft into training data and policy design. Advisor: Maj. Kullen Waggoner, Ph.D.
- **Relevant coursework:** Reinforcement Learning, Artificial Intelligence, Spacecraft Maneuver & Rendezvous, Orbit Determination (Kalman Filtering), Space Surveillance; sequences in Space AI, Astrodynamics, Space Systems, SSA
- **Honors:** Elected to Tau Beta Pi, The Engineering Honor Society (2026)

**Turkish Air Force Academy, Türkiye** 2017 – 2021
***B.S., Aerospace and Astronautical Engineering — GPA 3.82/4.0***

## RESEARCH EXPERIENCE

**Air Force Institute of Technology (AFIT), Wright-Patterson AFB, OH** Aug 2025 – Present
***Graduate Researcher — Space Autonomy & Reinforcement Learning***

**RL for SSA Sensor-Network Tasking (M.S. Thesis)**

- Formulating SSA sensor-to-target assignment as a sequential decision problem and extending deep-RL tasking policies to dynamic catalogs with maneuvering resident space objects (Python, MATLAB).

**Robustness of RL Sensor Tasking to Unpredicted Maneuvers**

- Built a maneuver-injection module that perturbs truth trajectories while leaving the estimator unaware, and evaluated a pretrained PPO tasking policy zero-shot against a greedy heuristic (3 sensors, ~100 objects) across maneuver rates.
- Measured tail failures, not just averages: 90th-percentile covariance, fraction of maneuvering objects left unobserved, and reacquisition time (Ray RLlib, TensorFlow).

**RL-Based Autonomous Spacecraft Docking**

- Formulated offset-port docking as an MDP with fully coupled 6-DOF rotational–translational dynamics under J2, capturing attitude–position coupling neglected by decoupled Clohessy–Wiltshire models.
- Built a custom Gymnasium environment with docking-corridor constraints, thruster limits, and structured reward shaping; ran systematic hyperparameter sweeps tracked in Weights & Biases.
- Trained PPO agents (PyTorch) that dock with tumbling targets up to 2.5 deg/s, where a classical PD baseline fails beyond ~1 deg/s, and quantified the fuel trade (~3–3.6 vs. 1–2 m/s Δv).

**Multi-Agent Cooperative Spacecraft Maneuver Planning**

- Developed a MAPPO framework (centralized training, decentralized execution) for multi-spacecraft proximity operations under HCW dynamics, benchmarked against an LQR + artificial-potential-field baseline.
- Accelerated convergence with a behavior-cloning warm start; diagnosed and resolved a mid-training policy collapse, reaching 100% task success at the final checkpoint; validated with Wilson score intervals, McNemar tests, and bootstrap CIs.

## PROFESSIONAL EXPERIENCE

**Turkish Air Force, Türkiye** Aug 2021 – Present
***Aerospace Engineer***

- UAV Pilot (Mar 2023 – Present), 1,000+ flight hours in operational environments; Pilot (2021 – 2023).

## PUBLICATIONS & PRESENTATIONS

- **Aytekin, O.**, Waggoner, K., Bagci, M., Avci, A., "Reinforcement Learning for Offset-Port Autonomous Docking with Coupled Rotational–Translational Dynamics in Low Earth Orbit," *77th International Astronautical Congress (IAC)*, Antalya, Türkiye, Oct 2026. Oral presentation.
- **Aytekin, O.**, Waggoner, K., Ergin, N. E., "Learning-Based Cooperative Maneuver Planning for Multi-Spacecraft Proximity Operations," *IEEE National Aerospace & Electronics Conference (NAECON)*, Cincinnati, OH, Aug 2026. DOI: 10.1109/NAECON70028.2026.11674936

## SKILLS

- **Machine Learning:** Deep RL (PPO, SAC, MAPPO), multi-agent RL (CTDE), behavior cloning, reward shaping, statistical evaluation; PyTorch, TensorFlow, Ray RLlib, Gymnasium, Stable-Baselines3, Weights & Biases
- **Programming:** Python, MATLAB, Git/GitHub, LaTeX
- **Aerospace:** HCW relative motion, coupled 6-DOF dynamics, J2 perturbations, orbit determination, Kalman filtering, SSA/SDA, rendezvous & proximity ops, attitude control
- **Languages:** Turkish (native), English (professional working proficiency) | **Interests:** Snowboarding, squash